

The effect of bidialectalism on executive function

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Abstract

Aims and objectives: We investigate how varying usage patterns in speakers of closely related language varieties might impact executive function. More specifically, bidialectals with more balanced usage were predicted to show better inhibitory control than less balanced bidialectals.

Design: Thirty-four adult bidialectals of Standard German and Swabian German performed two executive function tasks (flanker and Simon).

Data and analysis: The participants' reaction times on the two executive function tasks were analysed using regression models. Data and R code are available online.

Findings: Contrary to predictions, Swabian-dominant bidialectals showed smaller flanker and Simon effects than balanced German-Swabian bidialectals. Furthermore, contrary to some previous studies, executive function task performances correlated significantly.

Originality: We discuss how bidialectal language usage patterns can be assessed and how arbitrary analytical decisions affect findings regarding the effects of bidialectalism on executive function.

Significance: These findings shed a new light on the effects of bilingualism/bidialectalism on executive function.

Keywords

Bidialectalism, bilingualism, executive function, flanker task, Simon task

Using two (or more) languages has been linked to advantages in executive function (EF) compared to using only one language in daily life (see reviews by Adesope, Lavin, Thompson, & Ungerleider, 2010; Bialystok, Craik, Green, & Gollan, 2009; Kroll & Bialystok, 2013). However, a number of recent null findings has sparked a debate on whether previously found between-group differences may be grounded in factors other than bilingualism. Such factors relate to, for example, socio-economic

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status and relative language proficiency (Hilchey & Klein, 2011; Paap, Sawi, Dalibar, Darrow, & Johnson, 2013). The resulting question is thus under which conditions enhanced EF task performance emerges and which language environments are conducive for such effects to take place (Baum & Titone, 2014). It has been suggested that the grouping of bilinguals may have obscured within-group differences pertaining to relative usage of each language and the domains in which these languages were used, as well as the extent of language switching in varying contexts. A very specific group of speakers who use more than one language regularly are bidialectals, who are fluent in a standard language variety and a regional dialect. Bidialectal control of two typologically very close languages may differentially influence how much cognitive control needs to be exerted compared to bilinguals who speak two typologically more distinct languages. More specifically, bidialectals may have to exert differential cognitive control throughout their daily lives due to the ubiquitous usage of both dialects in their environment compared to bilinguals who may display a more compartmentalized language usage pattern. In order to explore whether navigating two languages regularly has effects on cognitive control, the focus of the present study is on highly proficient bidialectals who use their dialects to varying degrees in their daily lives. The within-language design furthermore avoids the necessity to meticulously match background variables across groups (Poarch & van Hell, 2017).

Differences between bilinguals and monolinguals in executive function

In their unity/diversity framework, Miyake and Friedman (2012) describe the so-called EF network. This cognitive control network is responsible for individuals' nonverbal cognitive processes and consists of the sub-components inhibition, tasks shifting and updating. This EF network is assumed to be influenced by sustained bilingual language control in that the repeated regulation of language competition in bilinguals affects those areas in the brain in which nonverbal EF processes take place (Abutalebi & Green, 2016; Green & Abutalebi, 2013).

In order to test individuals' EF, various nonverbal tests have been developed in the past. Two of these, commonly used to tap participants' ability to monitor the tasks and to inhibit interfering stimuli, are the Simon task (Simon & Rudell, 1967) and the flanker task (Eriksen & Eriksen, 1974). In the Simon task, it is the spatial position of the stimuli on the screen that is either congruent or incongruent with the required manual response. In the flanker task, the target stimulus is positioned centrally on the screen and the flanking stimuli are either congruent with the target or incongruent. Both these tasks have been employed in research comparing groups of monolinguals and bilinguals. In the literature, evidence for bilinguals outperforming monolinguals has been reported across the lifespan in studies with children (Carlson & Meltzoff, 2008; Poarch & Bialystok, 2015; Poarch & van Hell, 2012), young adults (Blumenfeld & Marian, 2014; Costa, Hernández, Costa-Faidella, & Sebastián-Gallés, 2009) and older adults (Bialystok, Craik, & Luk, 2008; Salvatierra & Rosselli, 2011).

Such differences have emerged in two ways: either bilinguals were faster in both the congruent and incongruent task conditions, which would indicate a monitoring advantage over monolinguals (e.g., Bialystok, Martin, & Viswanathan, 2005; Costa et al., 2009), or bilinguals performed equally on the congruent condition but were faster in the incongruent condition, which would indicate a more developed capacity to resolve conflict (and thus better inhibitory control) than monolinguals (Pelham & Abrams, 2014; Poarch & Bialystok, 2015; Poarch & van Hell, 2012). Both of these outcomes can be explained by the sustained practising of specific cognitive processes underlying bilingualism: nonverbal monitoring could be positively influenced by the need to monitor and switch between two languages, and nonverbal inhibitory control could be positively influenced by the need to repeatedly inhibit intrusion from the non-target language (see Hilchey & Klein, 2011, for a review).

However, differences are not always found in group comparisons (e.g., Antón et al., 2014; Gathercole et al., 2014; Paap & Greenberg, 2013), which has triggered a debate as to under which conditions differences between groups do or do not emerge (see Baum & Titone, 2014; Poarch & van Hell, 2017; Valian, 2015; van Hell & Poarch, 2014, for discussions). One of the factors that may be decisive in inducing differences in EF task performance in bilinguals compared to monolinguals is the pattern of bilinguals' language usage. More precisely, the frequency of using both languages on a daily basis and/or how often contexts demand switching between languages may be responsible for the effects on EF processing showing in some bilinguals and not in others (Blom, Boerma, Bosma, Cornips, & Everaert, 2017; Ross & Melinger, 2017). As pointed out above, bidialectals' language usage patterns may require differential cognitive control than bilinguals due to the possibly differing necessity and opportunity in daily life to use both languages. Thus, testing bidialectals may offer valuable insights into the frequency of using each language, of switching between the two languages, and its effect on EF, given their possibly broader spectrum of usage contexts compared to bilinguals. Furthermore, bidialectals, similarly to bilinguals who speak two distinct languages, need to control and switch between two language systems, which requires some form of EF. As Green and Abutalebi (2013) suggested in their Adaptive Control Hypothesis, varying dual-language interactional contexts (i.e., single-language, dual-language, or dense code-switching contexts) pose varying demands on language control processes. Accordingly, for example, bidialectals who experience more dual-language contexts "in which both languages are used but typically with different speakers" (p. 4) than bidialectals who experience more single-language contexts "in which one language is used in one environment and the other in a second environment" (p. 4) will also differ on how much inhibitory control they need to exert to communicate successfully. More specifically, dual-language contexts are assumed to require more inhibition than single-language contexts (see specific predictions below). Interest in bidialectals and EF has recently been sparked as evidenced by a small but growing body of studies that we now turn to.

Bidialectalism and EF

To our knowledge, five studies have so far explored EF in bidialectals, three in children and two in adults. Antoniou, Grohmann, Kambanaros, and Katsos (2016) compared children who were Cypriot Greek-Standard Modern Greek bidialectals, English-Greek bilinguals or Standard Modern Greek monolinguals. After statistically controlling for the bilinguals' and bidialectals' lower language proficiency in Greek compared to that of the Greek monolinguals, the authors found that the bilinguals and bidialectals outperformed the monolingual children in the Simon task (amongst other EF tasks). Furthermore, Ross and Melinger (2017) used the Simon task and the flanker task to compare bidialectal, bilingual and monolingual children. The results yielded no differences across groups save for higher accuracy for bilingual children than for the other two groups in the Simon task. The authors stress that their findings need to be interpreted tentatively and, critically, against the backdrop of dialect similarity as one of the factors that may play an important role in whether using two languages affects EF. This is all the more so relevant because "the cognitive implications of bidialectalism are underexplored" (p. 14) and "the question of bidialectalism has potentially important consequences for qualification of the bilingual advantage" (p. 15). Finally, Blom et al. (2017) tested four groups of children who were monolingual Dutch, bilingual Frisian-Dutch, bilingual Polish-Dutch or bidialectal Limburgish-Dutch. The children performed several EF tasks, one of which being the flanker task, on which they did not differ significantly. Significant differences did emerge, however, in the other EF task measuring selective attention. Critically, the effect of bilingualism in the Frisian-Dutch children was more robust than that in the Limburgish-Dutch bidialectals. The authors assume that the children's EF task performance was modulated by the differing language usage patterns between groups, such

that, for example, only Frisian-Dutch children receive Frisian instruction at school, while Limburgish-Dutch bidialectals have been found to code-mix more frequently.

At the other end of the age spectrum, Kirk, Fiala, Scott-Brown, and Kempe (2014) compared Simon task performance in older adults who were Gaelic-English bilinguals, Asian Language-English bilinguals, Dundonian Scots-English bidialectals or English monolinguals. The results showed that the four groups' performances did not differ, which the authors took to suggest that neither cultural and ethnic background attributed to inducing any EF differences across groups nor did the regular use of dialect in the case of the Dundonian Scots-English bidialectals. Finally, Scaltritti, Peressotti, and Miozzo (2017) compared Italian-Venetian speakers and Italian monolinguals on a flanker task. No differences were found across groups and bidialectals' estimated familiarity with dialect was not related to EF task performance. The authors interpret their findings as indicative of fewer opportunities for their bidialectal participants for language switching, particularly compared to the Catalan-Spanish speakers reported in the research by Costa et al. (2009), who outperformed the Spanish monolinguals in the EF task.

Bilingualism and bidialectalism: The Swabian situation

The particular bidialectalism investigated in this paper can be categorized as diaglossia, following Auer (2011, 487). According to Auer, and in contrast to diglossia, this situation is the most widespread in Europe today and is characterized by a continuum between a base dialect, intermediate varieties that carry a variable number of regional markers and the roofing standard language (p. 491). The general tendency, diachronically, is that the stronger dialectal forms are gradually being replaced in an increasing number of contexts of use by an increasing number of speakers in favour of more standard-like or "regiolectal" (p. 491) features. Due to space limitations, we refrain from giving a detailed account of the dialectal and regiolectal features of the Alemannic varieties currently categorized as Swabian. The most salient features of varieties of Swabian, including those that are intermediate between local base dialects and Standard German, are the palatalization of /s/ when followed by /t/ to /ʃ/, the lenization of Standard High German fortis consonants and, at least in base dialect, a considerable number of heteronyms not shared with the standard language (see Lameli, 2014; Ruoff, 1983, for dialectological details on this South-Western German variety). Given the diaglossic nature of the situation, the linguistic usage patterns referred to by the label Swabian are highly variable; they range from prototypical base dialect varieties to regiolectal sub-standard forms with much fewer features that differentiate the variety from the varieties categorized as standard language.

The particularity of bidialectalism, where dialect is first and foremost used in oral and informal contexts, entails certain adaptations in the design, scoring and interpretation of the questionnaire, to which we will return in the *Discussion* below.

The present study

The present study builds on and extends previous research on bilingualism and EF to a group of Swabian-German bidialectals to assess their language dominance/balance between dialect and standard language and the possible effect of this dominance/balance on their EF networks. The more balanced bidialectals were expected to show a smaller response time difference between the incongruent and the congruent conditions than less balanced bidialectals. This prediction is based on the assumption that incongruent conditions in EF tasks require resolving conflict induced by interfering information (Simon task = spatial position; flanker task = flanker direction) and that more balanced bidialectals should have accrued more practice in dealing with and resolving

language conflict than less balanced bidialectals. Bilingual populations' language usage patterns have been described using Green and Abutalebi's (2013) Adaptive Control Hypothesis in which language usage contexts pose varying demands on language control processes and will thus impact EF differently. In this view, dual-language contexts are assumed to require higher cognitive control demands in the form of inhibition than single-language and dense code-switching contexts, which are associated with lower cognitive demands. However, the bidialectal population in the present study experienced a mix of single- and dual-language contexts in their daily lives, so language contexts were not used to formulate the predictions above.

Furthermore, while this was not our objective at the outset of this study, a final goal of this report concerns the multitude of analytical choices that researchers working on the cognitive effects of bilingualism are faced with. Such choices can reasonably be settled in a number of ways, often with conflicting precedents in the literature, and are in that sense arbitrary. Below we highlight these arbitrary choices and explore how strongly they can affect the conclusions of studies such as ours.

Method

Participants

The participant sample consisted of 35 young adults (15 female, 20 male) who live in the German federal state of Baden-Württemberg in the greater Tübingen area, in which the regional dialect Swabian is regularly spoken. The participants received money for their participation. One participant claimed never to have learnt Swabian and was excluded from the analyses reported below, leaving a sample size of 34 participants (see Table 1).

The category of bidialectals is based on self-reports of the informants. We did not operationalize bilingualism/bidialectalism as a binary category, but as a radial category with several dimensions. For this purpose, participants completed an abbreviated form of the Bilingual Language Profile (BLP; Birdsong, Gertken, & Amengual, 2012; see online appendix, <http://doi.org/10.6084/m9.figshare.5379448.v1>, Section 7.1). While the original version of the BLP for German speakers assumes English to be the second language of participants, the abbreviated form of the BLP used in the present study replaced English with Swabian German for each of the questions. This questionnaire is an instrument that was developed to measure four different components that are deemed to be related to language dominance, or more generally to the role a language plays in the bilingual or multilingual repertoire. The dimensions cover language learning/usage history, current use, proficiency and attitudes. Our informants filled in the questionnaire providing information on the first three of the four dimensions for the Standard High German and Swabian components of the linguistic repertoire. As indicated above, the BLP uses self-reports to assess language dominance. Self-reports have been found to be a reliable measure of language proficiency (see, e.g., Gertken, Amengual, & Birdsong, 2014; Marian, Blumenfeld, & Kaushanskaya, 2007). While there are alternative instruments available (e.g., LEAP-Q, Marian et al., 2007), the BLP was chosen due to its relative brevity, efficiency and ease of administration both for the experimenter and for participants (see Gertken et al., 2014, p. 213). We did not collect responses on the participants' attitudes, given the overwhelmingly positive attitudes towards the Swabian dialect in this part of Germany (see the federal state of Baden-Württemberg's official ad campaign: "Wir können alles. Außer Hochdeutsch." [We can do everything. Except speak standard German.], Staatsministerium Baden-Württemberg, 2017), there was little additional information to glean from assessing the participants' attitudes on their dialect.

Self-rated proficiency, indicated on scales from "very poorly" (0) to "very good" (6), yielded mean proficiencies for Standard German ($M = 5.2$; $SD = 0.8$) and Swabian ($M = 5.0$; $SD = 1.2$). Furthermore, participants indicated their mean ages of acquisition (AoAs) for Standard German

Table 1. Mean scores (standard deviations) of participant characteristics.

Characteristic	Mean	SD
<i>n</i>	34 (15F/19M)	
Age (years)	22.6	4.1
Raven's raw score (shortened test, maximum possible: 18)	12.6	3.1
AoA Standard German (years)	3.5	3.3
AoA Swabian German (years)	2.5	4.1
Self-ratings ^a :		
Standard German proficiency (0–6) (average for speaking, listening, writing and reading)	5.2	0.8
Swabian German proficiency (0–6) (average for speaking and listening)	5.0	1.2
Usage ^b German	33%	27%
Usage ^b Swabian	61%	31%

^aSelf-report ratings of comprehension and fluency collapsed into proficiency score ranging from 0 to 6, where 0 was “very poor” and 6 was “very good”.

^bUsage consists of collapsed self-reports of daily language usage (out of 100%) with friends, family and at university/work.

AoA: age of acquisition.

($M = 3.5$ years; $SD = 3.3$) and Swabian ($M = 2.5$ years; $SD = 4.1$). Finally, relative daily usage with friends, family and at work was indicated on scales from 0% to 100% for Standard German ($M = 33\%$; $SD = 27\%$) and Swabian ($M = 61\%$; $SD = 31\%$). Missing data for one variable (*At what age did you start to feel comfortable using Swabian?*) were replaced by the highest possible value (see supplementary materials, Section 7.1, for more details). Furthermore, all responses to the question “*How many years of education did you have in Swabian?*” were set to 0, since Swabian-medium education does not exist (three participants provided non-zero responses to this question).

Certain items in the original questionnaire were found to have a problematic fit to a diaglossic situation. Nevertheless, we deem the questionnaire data, if interpreted adequately, to be a suitable means of tapping into the weight dialectal varieties have in the bidialectal repertoires. Unlike in settings that may seem more typical for societal bilingualism, in a diaglossic situation the mastery, status and use of the standard language is undisputed. Thus, in our analyses below, a particular focus will lie on the scores of the dialect component, which allows us to assess the importance of Swabian in the participants’ language history and use as well as their self-assessed proficiency in dialectal varieties.

On the basis of the participants’ responses, Swabian dominance scores were computed using the guidelines at <https://sites.la.utexas.edu/bilingual/scoring-and-interpreting-the-results/> (see also the online appendix, Section 7.1). Positive values suggest the participants are dominant in Swabian, negative values that they are dominant in Standard German and values close to zero suggest balanced bidialectalism.

Tasks

Background measure. The participants completed a test of nonverbal fluid intelligence (the odd-numbered puzzles of Raven’s Advanced Progressive Matrices, Set II; Raven, Raven, & Court, 1998). This measure served a purely descriptive purpose and was not included in the inferential analyses as it is conceivably influenced in part by the cognitive and sociological consequences of bidialectalism¹ (on “post-treatment variables”, see Huitema, 2011, Chapter 8).

Executive function. Participants performed two EF tasks: the flanker task and the Simon task, both of which were computer-administered.

Flanker task. In the flanker task, participants see an array of five arrows on the screen: a central target arrow that is flanked on the right and the left by two non-target arrows that point either in the same direction as the central arrow or in the opposite direction. Participants use the left or right button of a response box to indicate the direction of the central target arrow. There are two types of trials: congruent trials with flankers pointing in the same direction as the target, and incongruent trials with flankers pointing in the opposite direction as the target. There is either no conflict (congruent trial) or there is conflict (incongruent trial) that needs to be resolved. Participants were shown 192 pictures with five left- or right-pointing arrows each. There were 48 trials for each combination (left/right, congruent/incongruent). The trials appeared in a new random order for each participant. Each trial was prompted by a “+” sign appearing in the centre of the screen for 800 ms followed by a 250 ms pause. The arrows remained on-screen until a response was given. The intertrial interval lasted 500 ms. After 64 and 128 trials, the participants were able to take a short self-timed break. The “blocks” that these pauses defined served no purpose in our analysis of the data. Before the actual task, participants completed an eight-stimulus training run that they had to redo if they made any error. No feedback was given during the actual task.

Simon task. In the Simon task, red or blue squares appear individually after a fixation cue on a screen and participants press the left or right button of a serial response box to indicate which colour they are viewing. The squares appear either left of fixation or right of fixation. There are two types of trials: congruent trials in which, for example, a red-colour square appears on the right of the screen requiring a right button press, and incongruent trials in which, for example, a red-colour square appears on the left requiring a right button press. Incongruent trials induce spatial conflict in participants, which requires inhibitory processes for correct task performance. This task consisted of 192 trials in which either a blue or a red rectangle appeared at either the left- or right-hand side of the screen. The participants were asked to press the left button when a red rectangle appeared and the right button when a blue rectangle appeared. On congruent trials, the position of appearance matched the position of the correct response; on incongruent trials, it did not. There were 48 trials for each combination (blue/red, congruent/incongruent). The trials appeared in a new random order for each participant. Each trial was prompted by a “+” sign appearing in the centre of the screen for 800 ms followed by a 250 ms pause. The blue or red rectangle remained on-screen until a response was given. The intertrial interval lasted 500 ms. After 64 and 128 trials, the participants were able to take a short self-timed break. The “blocks” thus defined serve no analytical purpose. Before the actual task, participants completed an eight-stimulus training run that they had to repeat if they made any error. No feedback was given during the actual task.

Procedure

The participants first signed an informed consent form, then filled out the language background questionnaire, after which they were administered the Raven’s task. Finally, participants were randomly assigned to complete the flanker task or the Simon task first. At the time of testing, research with human participants did not undergo ethical board review at the Faculty of Humanities at the University of Tübingen.

Table 2. Participants’ mean accuracy rates per participant by task and condition.

Task	Condition	Mean acc in (%)	SD (%)	Minimum (%)	Maximum (%)
Flanker	Congruent	99.8	0.6	97.9	100
	Incongruent	97.9	2.1	92.7	100
Simon	Congruent	98.0	1.6	94.8	100
	Incongruent	97.5	2.6	87.5	100

SD: standard deviation.

Results

We present three analyses of the flanker and Simon data. In the first analysis, we assess our prediction at the outset of the study, namely that more balanced bidialectals should show smaller magnitudes of flanker and Simon effects. Then, we investigate a plausible alternative hypothesis that only occurred to us after the study. Lastly, cognizant of the many ways of analysing these data that would still fit in with the overarching question of whether bidialectalism is associated with enhanced cognitive function, we explore how strongly arbitrary analytical choices affect the results. All data and R code are available from the online appendix.

Data transformation and exclusion

As is typical for response latencies, the individual response latencies on both the flanker and Simon task were positively skewed. For this reason, we transformed the latencies (e.g., 333 ms per stimulus) to response speeds (e.g., 3 stimuli per second) by inverting the latencies and multiplying the results by 1000. The distributions of these response speeds were approximately bell-shaped. Alternative transformations are considered at the end of the *Results* section.

Since we did not decide on an outlier exclusion criterion beforehand, no data points were removed for this reason in the two main analyses. The analysis at the end of the *Results* section also considers the effect of several common outlier exclusion criteria on the results.

Participants made very few errors in the Simon and flanker tasks (Table 2). Furthermore, we did not observe a relationship between the participants’ accuracy scores and their response speeds in the congruent and incongruent Simon and flanker conditions that would hint at a speed–accuracy trade-off (see the online appendix, Section 6.2.1). Since we did not decide beforehand to exclude incorrect responses from the analyses, they are retained in the analyses reported here. The analysis at the end of the *Results* section considers the effect of excluding incorrect responses.

Finally, flanker and Simon effects were computed for each participant by subtracting their mean response speed in the incongruent condition from their mean response speed in the congruent condition (see Table 3). Overall, 33 out of 34 participants showed the expected positive flanker effect, and 28 showed the expected positive Simon effect. These flanker and Simon effects were correlated at $r = 0.46$. The analysis below explores the effect of different expressions of the flanker and Simon effects on the results.

Planned analysis: Smaller effect magnitudes for balanced bidialectals

Our hypothesis at the outset was that balanced bidialectalism would be associated with enhanced cognitive function. Put differently, the magnitude of the flanker and Simon effects should be lowest

Table 3. Participants’ mean response speed (stimuli/stimulus) by task and condition as well as magnitude of flanker and Simon effects.

Task	Condition	Mean response speed	SD	Minimum	Maximum
Flanker	Congruent	2.49	0.29	1.87	3.15
	Incongruent	2.27	0.28	1.72	2.92
	Flanker effect	0.23	0.11	−0.02	0.49
Simon	Congruent	2.66	0.33	1.94	3.32
	Incongruent	2.52	0.31	1.97	3.13
	Simon effect	0.13	0.12	−0.08	0.39

Note. Flanker/Simon effects calculated by subtracting incongruent from congruent response speeds.

for bidialectals with dominance scores around 0. This implies that the relationship between the participants’ bidialectal dominance score and their flanker/Simon effects should be nonlinear. We therefore allowed for the effect of bidialectal dominance on the flanker/Simon effects to be potentially nonlinear in generalized additive models. However, as Figure 1(a) shows, the effect of bidialectal dominance was estimated to be linear such that Swabian-dominant bidialectals showed the smallest flanker effects ($\hat{\beta} \pm \text{SE} = -0.0008 \pm 0.0004$, $t(32) = 2.3$, $p = 0.03$). This linear trend, while significant, is not consistent with our prediction of a nonlinear effect. For the Simon task (Figure 1(b)), the trend was similarly estimated to be linear, again contrary to our prediction ($\hat{\beta} \pm \text{SE} = -0.0009 \pm 0.0004$, $t(32) = 2.4$, $p = 0.02$).

For the sake of completeness, Figures 1(c) and (d) show an alternative way of testing this prediction. For these analyses, the absolute values of the bidialectal dominance variable were used as the predictor, the expectation being that the flanker/Simon effects are smaller for lower predictor values as these are indicative of more balanced bidialectalism. For neither the flanker nor the Simon data did we find evidence that supports this prediction (flanker: $\hat{\beta} \pm \text{SE} = -0.0001 \pm 0.0007$, $t(32) = 0.2$, $p = 0.85$; Simon: $\hat{\beta} \pm \text{SE} = 0.0006 \pm 0.0007$, $t(32) = 0.8$, $p = 0.43$). Mixed-effects models for which the individual response speeds were not aggregated first point to the same conclusions. These models are reported in full in the online appendix (Section 10.1-2).

Post-hoc analysis: Smaller flanker/Simon effects for greater contribution of Swabian to bidialectal repertoire

In contrast to our initial hypothesis that balanced bidialectalism enhances cognitive function, an alternative view of the language constellation and language usage patterns for those living in the Southern German federal state of Baden-Württemberg could be that Standard German is the majority language most used in the media and in schools and Swabian German is the home language for many. Hence, bidialectals will not be able to avoid processing Standard German in their daily lives. Accordingly, in contrast to our initial prediction formulated above, we posited that the sole BLP’s Swabian score instead of the overall language balance might better indicate how bidialectal the participants were. This in turn would mean that those speakers who displayed the highest Swabian scores would be considered the more balanced bidialectals, who then would also be predicted to show enhanced inhibitory control compare to speakers with lower Swabian proficiency. In a post-hoc analysis of the Swabian score only, we explored whether participants with high total Swabian scores displayed smaller magnitudes of the flanker/Simon effects. We modelled this relationship, allowing for the effect of total Swabian score on the flanker/Simon effects to be potentially nonlinear in generalized additive models.

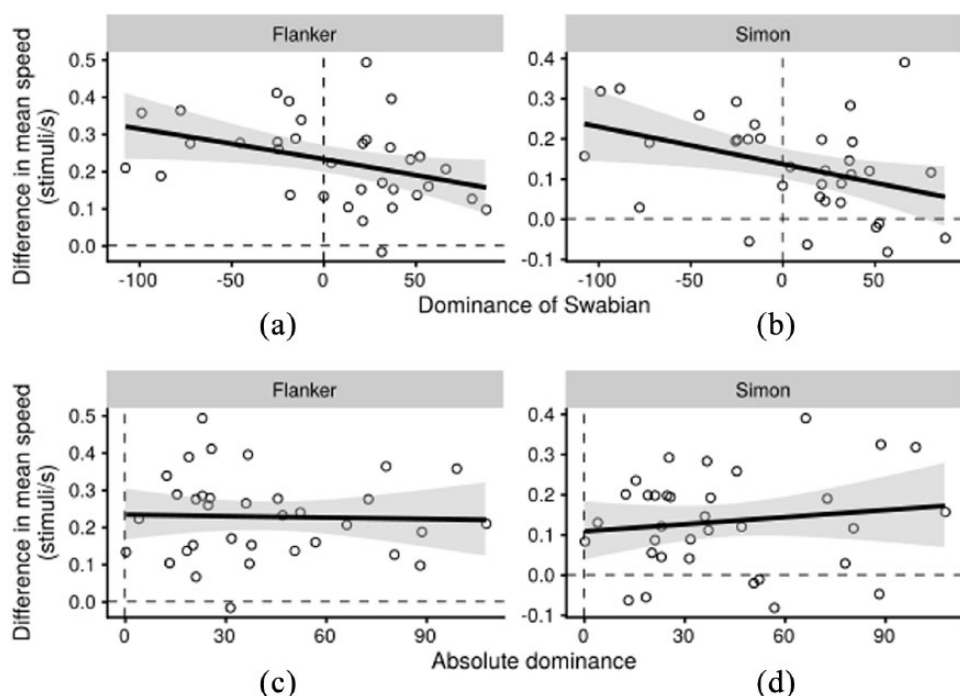


Figure 1. Flanker and Simon effects according to bidialectal dominance in Swabian. In all plots, the scatterplot smoothers were allowed to be nonlinear, but all produced linear fits. Shaded regions are 95% confidence bands. (a), (b) Flanker and Simon effects by raw dominance in Swabian. Dominance scores around 0 indicate balanced bidialectalism; positive values indicate dominance in Swabian, negative values dominance in standard German. (c), (d) Flanker and Simon effects by absolute dominance. Lower dominance scores close to 0 indicate more balanced bidialectalism.

As Figure 2 shows, the effect of total Swabian was estimated to be linear such that Swabian-dominant bidialectals showed the smallest flanker/Simon effects. For the flanker task, this linear trend was not significant ($\hat{\beta} \pm \text{SE} = -0.0010 \pm 0.0006$, $t(32) = 1.7$, $p = 0.10$), while for the Simon task it was ($\hat{\beta} \pm \text{SE} = -0.0013 \pm 0.0006$, $t(32) = 2.2$, $p = 0.03$).

Mixed-effects models for which the individual response speeds were not aggregated first point to the same conclusions. These models are reported in full in the online appendix (Section 10.3).

Exploring the effects of arbitrary analytical choices in a multiverse analysis

For the analyses reported above, we took a number of decisions that were arbitrary in the sense that different analysts could reasonably make choices different from ours when analysing the same data. Firstly, we did not exclude latencies to incorrect responses. While some studies on the effects of bilingualism on EF did not report to have excluded latencies for incorrect responses (e.g., Bialystok, Craik, & Luk, 2008; Bialystok, Martin, & Viswanathan, 2005), others did (e.g., Luk, de Sa, & Bialystok, 2011). Secondly, we did not exclude very short response latencies that may be the result of accidental button presses, and even had we done so, the decision where to put the threshold would have been arbitrary in itself as different thresholds

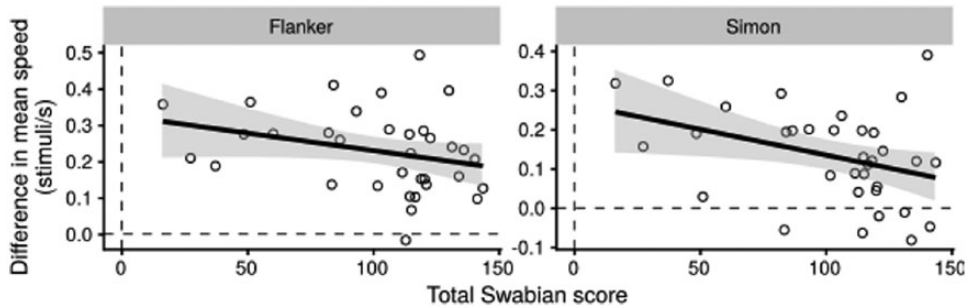


Figure 2. Flanker and Simon effects according to the participants' total Swabian score. The scatterplot smoothers were allowed to be nonlinear, but all produced linear fits. Shaded regions are 95% confidence bands.

are used in the literature (e.g., latencies shorter than 200 ms in Poarch & van Hell, 2012; 250 ms in Bialystok & DePape, 2009; 300 ms in Keijzer & Schmid, 2016). Thirdly, we transformed the response latencies to response speeds, which substantially alleviates the right skew in the outcome distribution, but to our knowledge, most related studies do not discuss this point and use raw response latencies (see also Zhou & Krott, 2016, for differences between bilinguals and monolinguals in EF task response latencies in the tail of the distribution). A reasonable alternative would be to logarithmically transform the response latencies. Fourthly, we did not exclude responses more than a given number of standard deviations from the participants' mean (as in, e.g., Paap & Greenberg, 2013) or from the overall mean (as in Kirk et al., 2014). Fifthly and lastly, we expressed the magnitude of the flanker and Simon effects as differences between the congruent and incongruent condition means. Occasionally, however, these effects are expressed as ratios or proportional differences (e.g., a 7% increase in response latencies in the incongruent condition relative to the congruent condition), not as raw differences (see Bialystok & DePape, 2009; Kousaie & Phillips, 2012, albeit for Stroop task results).

The BLP-based predictor similarly permits substantial flexibility on the part of the researcher. We can cast the overarching hypothesis in terms of (a) raw dominance of Swabian (the more dominant in Swabian, the better cognitive control), (b) absolute dominance of Swabian (the more balanced, the better cognitive control) or (c) total contribution of Swabian to the bidialectal repertoire (the greater the contribution, the better cognitive control), as we have done in the analyses above. Other alternatives would be defensible, too, however. For instance, since the Swabian dialect is rarely written, we could reasonably ignore the BLP items pertaining to reading and writing and consider the effect of (d) raw dominance of Swabian, considering oral use and skills only, (e) absolute dominance of Swabian (oral only) and (f) total contribution of Swabian (oral only) to the bidialectal repertoire. Finally, some may consider that only the BLP's *use* dimension has any relevance in affecting cognitive control. Such a view yields two more possible expressions of the dependent variable: (g) raw dominance of Swabian in language use and (h) absolute dominance of Swabian in language use. Note that most of these expressions of the predictor variable are inconsistent with our present hypothesis, but that they are all reasonable under the more general hypothesis that bidialectalism affects EF.

When the flexibility in expressing the predictor variable is combined with the flexibility in preparing the flanker and Simon data, there are multiple ways in which these data could have been analysed. Most (although perhaps not all) of these would probably have made sense had

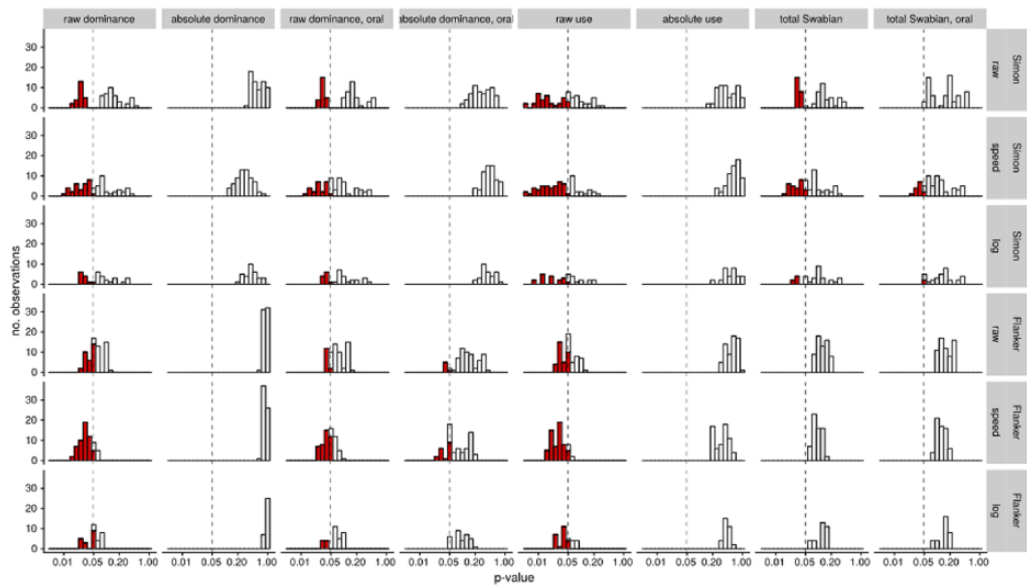


Figure 3. Results of a multiverse analysis on the flanker and Simon data. Different columns show the results for different expressions of the predictor variable (degree of bidialectalism), while different rows show the results for the different tasks (flanker versus Simon) and for different transformations of the latency data (raw latencies, transformed to speeds, logarithmically transformed). The variation within each panel is due to the data exclusion criteria and whether the flanker and Simon effects are expressed as differences or as ratios. The p -values are plotted along a logarithmically transformed axis. Significant p -values ($p < 0.05$) are coloured red (colour online only).

they produced inferential results consistent with one's expectation as to whether bidialectalism affects EF. To explore how strongly the results of this study vary due to such arbitrary decisions, we carried out a “multiverse” analysis (Steegen, Tuerlinckx, Gelman, & Vanpaemel, 2016). In a multiverse analysis, the data are analysed once for each sensible combination of arbitrary analytical choices, and the distribution of results (here: p -values) is reported in its entirety.

For the present study, we considered 2560 alternative ways of analysing the data. The full method and results are available in the online appendix (Section 11); Figure 3 presents the main results.² From these results, it emerges that the different ways to assess whether *balance* of bidialectalism is associated with flanker or Simon performance (predictor variables: absolute dominance; absolute dominance, oral; absolute use) all tended to yield non-significant results, irrespective of transformations and data exclusion criteria. Our null finding with respect to our original hypothesis, then, is clearly not due to arbitrary analytical decisions. Similarly, analyses that assess the effect of the contribution of Swabian to the bidialectal repertoire (predictor variables: total Swabian; total Swabian, oral) more often than not yielded insignificant results. Here, too, we conclude that our null result is not an artefact of arbitrary analytical decisions.

The independent variables found to be most associated with a significant effect on EF are raw dominance of use in Swabian, which means the extent to which the participants report using Swabian more often than Standard German, and raw dominance proper. However, this finding does not represent our initial prediction.

Discussion

Bidialectals were tested on two commonly used EF tasks: the flanker and the Simon task. The rationale for testing bidialectals was that it is as yet unclear whether the specific language usage patterns shown by bidialectals, in contrast to bilinguals, attenuate or modulate the cognitive demands of controlling the two languages differently. The need to control two typologically close languages ubiquitously used in their language environment may pose different language control demands than that necessary for bilinguals who speak two more typologically distinct languages used in more distinct domains. Consequently, testing bidialectals allows assessing whether varying usage patterns between closely related languages impact EF differentially. More specifically, we predicted that more balanced bidialectals would show better inhibitory skills compared to less balanced bidialectals. The results, however, did not confirm our predictions and yielded the smallest flanker and Simon effects for Swabian-dominant bidialectals. In the following, we discuss methodological considerations in terms of how data from these EF tasks can be analysed statistically and in terms of using the BLP as a measure to assess the language backgrounds of bidialectals.

The impact of reasonable but arbitrary analytical choices

It is rarely a priori clear how analytical issues such as whether and how to transform the data or to exclude certain data points are best settled, and precedents for different choices can be found in the literature. In this sense, many decisions taken in the course of any one analysis are arbitrary. A multiverse analysis – a type of robustness check – indicated that the majority of alternative analyses yielded results similar to our main analyses, that is, they did not confirm our predictions. However, the multiverse analysis also illustrates that, depending on how such decisions are resolved, evidence that dominance and use of Swabian is related to executive functioning could be found. In the light of conflicting findings about the effect of bilingualism on executive functioning, we think it would be useful if future studies in this field attempted either to minimize the effect arbitrary decisions can have on the studies' conclusions (e.g., by preregistering these decisions) or to gauge to what extent they are affected by them (e.g., through a multiverse analysis or by having the same dataset analysed by a number of independent teams; e.g., Silberzahn et al., 2017).

Use of the BLP instrument in the present study

While the BLP is a quick and relatively easy to administer measure to assess speakers' language dominance (see the *Method* section), we found the BLP to have a less than ideal fit to diaglossic situations. Dialectal varieties such as Swabian, to give but the most obvious example, are hardly ever officially used as a language of (literacy-based) education. The answers to the item "years of schooling in Swabian", therefore, had to be manually set to 0 even if some informants gave other answers. In retrospect, therefore, we are not entirely convinced that the particular diaglossic situation of many bidialectal speakers is captured adequately by an instrument such as the BLP; given the complementary distribution of the varieties with the standard form covering an increasing number of language use domains, the BLP items that presuppose a somewhat more typical situation – where two languages can potentially cover all domains and interactions – might entail important measurement errors, in particular in the language history component. Moreover, since dialect proficiency in literacy is hardly ever an issue (incidentally, a situation shared with many minority language situations where the minority

language's use in literacy is often scarce to absent), the proficiency scores, too, might be rather unreliable. In a diglossic relationship between the strong standard and the weak dialect, with standard-near patterns being the default choice in an increasing number of domains for an increasing number of speakers, it is clear that questionnaire items tapping into the exposure to and use of the weaker language are first and foremost a measure of the speakers' resistance to the general language shift. Thus, the inference seems to be that the higher these scores, the more bidialectal/bilingual the individual's reality is. As shown in the *Results* section, we can use the dialect scores as predictors for the weight of the dialect in the repertoire. However, this is a considerable deviation of the initial purpose of the BLP instrument and we are aware that this use of some sub-components of the instrument calls for further validation in independent studies with larger samples.

Conclusion

In conclusion, the bidialectals tested in the present study displayed performances on two EF tasks indicating that those who used the non-standard dialect to a greater degree in daily life showed better EF skills than those who used Standard German and the regional dialect in a more balanced manner. Hence, given the ubiquitous usage of the majority language (in this case, Standard German) outside the home and in the media, at least for bidialectals it is not so much a balanced usage of the two languages but the amount of exposure to and usage of the minority language (in this case, the regional dialect) that may have a greater impact on the development of EF than previously assumed. Thus, future research may need to allow for a more fine-grained assessment of exposure and usage patterns, particularly of the minority language of bilinguals and bidialectals. Such an assessment may also be helpful in better determining different levels of bilingualism and bidialectalism in the respective populations and in better explaining diverging results in EF task performance across studies.

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Notes

1. Pearson correlations: Raven's score and Flanker conflict effect ($r = -.10$); Raven's score and Simon effect ($r = -.31$).
2. It should be noted that even this multiverse analysis did not account for any flexibility we had in deciding how to deal with missing or patently false questionnaire responses.

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